6CCSACCM
Cooperation, Coordination and Multi-Agent Systems

Week 5

Cooperative games (part II)

Maria Polukarov

Beyond Transferable Utility

- So far, we assumed that the payoff earned by a coalition is allocated to the coalition as a whole and can be shared arbitrarily.
- Another possibility: payoffs accrue to individual coalition members and cannot be transferred.
- With no transfers, we have to reason about coalitional actions:
  - $\{x, y\}$ can perform a, which pays 1 to x and 6 to y
  - $\{x, y\}$ can perform b, which pays 4 to x and 2 to y

NTU game example: academic collaboration

- n researchers working at n different universities can form groups to collectively write papers.
- The composition of a group determines the quality of the paper they produce.
- Each author receives a payoff from their own university:
  - promotion;
  - bonus;
  - teaching load reduction;
  - a parking spot.
- Payoffs are *non-transferable*.

Hedonic games

- Special case—each coalition only has one action available to it:
  - i's payoff is determined by the coalition i is in.
- Such games are called **hedonic games**.
- In hedonic games, each player *ranks* all coalitions she could possibly join, from best to worst (ties are possible):
  - 1: $\{1,2\} \sim \{1,3\} \succ \{1,2,3\} \succ \{1\}$;
  - 2: $\{2\} \succ \{1,2,3\} \succ \{1,2\} \succ \{2,3\}$;
  - 3: $\{1,2,3\} \succ \{3\} \succ \{3,2\} \sim \{1,3\}$.

Hedonic games: formal definition

- A set of players $N = \{1, 2, \ldots, n\}$
- Players want to split into coalitions forming a *partition* of N
- Each player i has preferences $\succeq_i$ over $\{S \subseteq N | i \in S\}$ (all coalitions she could be in)
- Outcome: coalition structure (partition)
  - note: no payoff distribution
- Note: without loss of generality, one can encode preferences by numbers:
  - 1: $\{1, 2\} \sim \{1, 3\} \succ \{1, 2, 3\} \succ \{1\} \rightarrow$ $v_1(\{1, 2\}) = v_1(\{1, 3\}) = 2,\ v_1(\{1, 2, 3\}) = 1,\ v_1(\{1\}) = 0$

Compactly representable hedonic games

- Anonymous games: preferences depend on coalition sizes only.
- Each agent assigns a value to every other agent—$v_i(j) : N \setminus \{i\} \to \mathbb{R}$:
  - additively separable hedonic games:

$$v_i(S) = \sum_{i \in S \setminus \{i\}} v_i(j);$$

  - fractional hedonic games:

$$v_i(S) = \frac{\sum_{i \in S \setminus \{i\}} v_i(j)}{|S|};$$

  - B-games: $v_i(S) = \max_{i \in S \setminus \{i\}} v_i(j)$;
  - W-games: $v_i(S) = \min_{i \in S \setminus \{i\}} v_i(j)$.

Additively separable hedonic games vs. induced subgraph games

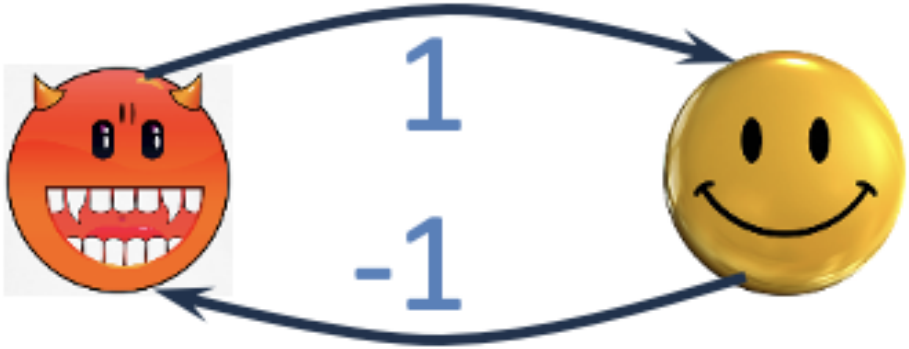
- Both games: players are vertices of a weighted graph.
- Induced subgraph games:
  - edges are undirected;
  - value of a coalition T = total weight of internal edges.
- ASHGs:
  - edges are directed;
  - value that i assigns to a coalition T = total weight of edges from i to players in $T \setminus \{i\}$.

Individual rationality

- A partition is **individually rational** if no player strictly prefers being on its own to its current coalition.
- Minimum stability requirement.

Nash stability

- A partition is said to be **Nash stable** if no player can benefit from moving from its coalition S to another coalition T.
- Formally, player i has an *NS-deviation* from a partition P if there is a coalition $T \in P \cup \{\emptyset\}$ such that i prefers $T \cup \{i\}$ to its current coalition.
  - A partition is Nash stable if no player has an NS-deviation.
- There is a two-player game with no Nash stable outcomes:



Nash stability: symmetric ASHGs

- **Symmetric ASHGs**: $v_i(j) = v_j(i)$.
- Observation: every such game admits a Nash stable outcome:
  - Every NS-deviation increases the sum of utilities.

Theorem (Gairing, Savani'10)

Finding a Nash stable outcome in symmetric additively separable hedonic games is PLS-complete.

Individual stability

- Player i has an *IS-deviation* from a partition P if there is a coalition $T \in P \cup \{\emptyset\}$ such that i prefers $T \cup \{i\}$ to its current coalition and all players in T weakly prefer $T \cup \{i\}$ to T.
  - A partition is individually stable if no player has an IS-deviation.

Contractual individual stability

- Player i has a *CIS-deviation* from a partition P if there is a coalition $T \in P \cup \{\emptyset\}$ such that:
  - i prefers $T \cup \{i\}$ to its current coalition S;
  - all players in T weakly prefer $T \cup \{i\}$ to T;
  - all players in S weakly prefer $S \setminus \{i\}$ to S.
- A partition is contractually individually stable if no player has a CIS-deviation.

Contractual individual stability

Theorem

Every hedonic game has a CIS outcome.

Proof.

- assume each player assigns a numerical utility to each coalition (we argued this is equivalent);
- then each CIS deviation increases the sum of utilities;
- thus, a partition that maximizes the sum of utilities is CIS.

□

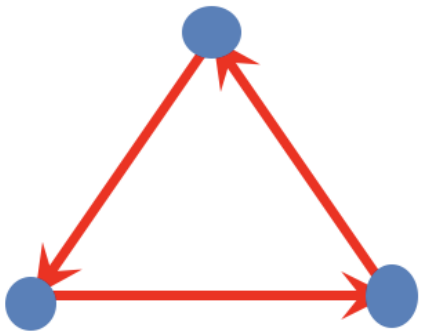
Core stability

- A group of players C *blocks* a partition P if each player in C prefers C to its coalition in P.
- A partition is **core stable** if no group blocks it.

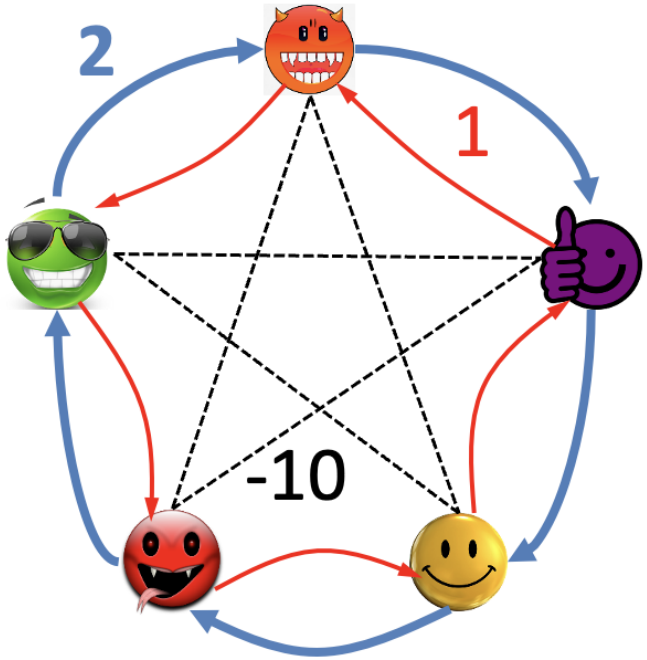
Core stability: empty core

Core may be empty:

- 1: $\{1,2\} \succ \{1,3\} \succ \{1\} \succ \{1,2,3\}$;
- 2: $\{2,3\} \succ \{2,1\} \succ \{2\} \succ \{1,2,3\}$;
- 3: $\{3,1\} \succ \{3,2\} \succ \{3\} \succ \{1,2,3\}$.



Additively separable game with an empty core



- 1: No coalition of size 3, 4, or 5 is IR.
- 2: No partition into coalitions of size 1 and 2 can be stable.
- 3: Slightly more complex construction for symmetric games.

Core stability: friends and enemies

Friends and **enemies** games—additively separable hedonic games where each player i partitions $N \setminus \{i\}$ as friends and enemies:

- some positive utility F for friends;
- some negative utility E for enemies;
- need not be symmetric.

Core stability: friends and enemies

- **Friend appreciation game:** $F = n$, $E = -1$
- **Enemy aversion game:** $F = 1$, $E = -n$

Theorem

Every friend appreciation game and every enemy aversion game has a non-empty core.

Stable matching problem

- Set X of women and set Y of men, $|X| = |Y| = n$.
- Each $x \in X$ has a preference order $\succ_x$ over all $y \in Y$.
- Each $y \in Y$ has a preference order $\succ_y$ over all $x \in X$.
- For every person being unmatched is least preferred.
- For matching M let $M(x) \in Y$ be the partner of woman $x \in X$ in M, and $M(y) \in X$ the partner of man $y \in Y$ in M.
- $M(x) = *$ if x unmatched in M, and $M(y) = *$ similarly.

Anna:	Jacob $\succ$ Isaac $\succ$ Ken	Isaac:	Anna $\succ$ Bella $\succ$ Clare
Bella:	Isaac $\succ$ Jacob $\succ$ Ken	Jacob:	Bella $\succ$ Anna $\succ$ Clare
Clare:	Isaac $\succ$ Jacob $\succ$ Ken	Ken:	Anna $\succ$ Bella $\succ$ Clare

Perfect matching

- A **matching** is a set M of ordered pairs $x - y$ with $x \in X$, $y \in Y$, such that
  - each woman $x \in X$ appears in at most one pair of M;
  - each man $y \in Y$ appears in at most one pair of M.
- A matching is **perfect** if $|M| = |X| = |Y| = n$.

Example: $M = \{A - K, B - J, C - I\}$

Anna:	Jacob $\succ$ Isaac $\succ$ **Ken**	Isaac:	Anna $\succ$ Bella $\succ$ **Clare**
Bella:	Isaac $\succ$ **Jacob** $\succ$ Ken	Jacob:	**Bella** $\succ$ Anna $\succ$ Clare
Clare:	**Isaac** $\succ$ Jacob $\succ$ Ken	Ken:	**Anna** $\succ$ Bella $\succ$ Clare

Blocking pair

- Given a perfect matching M, woman x and man y form a **blocking pair** if
  - x prefers y to her partner in M;
  - y prefers x to his partner in M.
- Key point: a blocking pair can jointly improve.

Example: Bella and Isaac form a blocking pair.

Anna:	Jacob $\succ$ Isaac $\succ$ **Ken**	Isaac:	Anna $\succ$ Bella $\succ$ **Clare**
Bella:	Isaac $\succ$ **Jacob** $\succ$ Ken	Jacob:	**Bella** $\succ$ Anna $\succ$ Clare
Clare:	**Isaac** $\succ$ Jacob $\succ$ Ken	Ken:	**Anna** $\succ$ Bella $\succ$ Clare

Stable matching

- A **stable matching** is a perfect matching with no blocking pairs.
- <u>Stable matching problem:</u> give the preference lists of n men and n women, find a stable matching (if one exists).

Example: $M = \{A - I, B - J, C - K\}$

Anna:	Jacob $\succ$ **Isaac** $\succ$ Ken	Isaac:	**Anna** $\succ$ Bella $\succ$ Clare
Bella:	Isaac $\succ$ **Jacob** $\succ$ Ken	Jacob:	**Bella** $\succ$ Anna $\succ$ Clare
Clare:	Isaac $\succ$ Jacob $\succ$ **Ken**	Ken:	Anna $\succ$ Bella $\succ$ **Clare**

Stable matching as a hedonic game

An instance of stable matching can be seen as a hedonic game:

- each man y prefers being alone to any coalition not of the form $\{x, y\}$, where x is a woman;
- he prefers every coalition of the form $\{x, y\}$ to being alone;
- his preferences over coalition of the form $\{x, y\}$ are determined by his preferences over X.

Stable matching = outcome in the Core

Stable roommate problem

- $2n$ people; each person ranks others from 1 to $2n-1$;
- assign roommate pairs so that there are no blocking pairs.

Example: no perfect matching is stable.

Adam:	B $\succ$ C $\succ$ D
Bob:	C $\succ$ A $\succ$ D
Chris:	A $\succ$ B $\succ$ D
Dan:	A $\succ$ B $\succ$ C

$\{A-D, B-C\}$: $A-C$ is a blocking pair;
$\{A-C, B-D\}$: $A-B$ is a blocking pair;
$\{A-B, C-D\}$: $B-C$ is a blocking pair.

Gale-Shapley deferred acceptance (DA) algorithm

Each women starts off being matched with $\emptyset$.
The man-proposing DA proceeds in rounds:

- Each man, y, makes a proposal to his most preferred woman. After this, each woman, x, that has received a proposal tentatively accepts the proposal from her most preferred man and rejects the rest.
- Each man, y, whose proposal was rejected in the previous round makes a proposal to his next most preferred woman. Second, each woman, x, that has a new proposal tentatively accepts the proposal from her most preferred man and rejects the rest.
- The DA algorithm terminates when no new proposals are made, at which point the matching that corresponds to proposals that are tentatively accepted is made final.

DA algorithm: termination

- Men propose to women in decreasing order of preference.
- Once a woman is matched, she never becomes unmatched; she only "trades up".

Proposition

DA terminates after at most $|X| \cdot |Y| = n^2$ *iterations.*

Proof.

- Each time a man proposes to a new woman;
- only n^2 possible proposals.

$\square$

DA algorithm: perfection

Proposition

After DA terminates, all men and all women are matched.

Proof.

- Suppose for the sake of contradiction that some man y remains unmatched.
- Then, since $|X| = |Y|$, some woman x remains unmatched.
- y has proposed to all women in his list—so, in particular, to x.
- Thus, x was matched at some point, with contradicts our previous observation.

$\square$

DA algorithm: stability

Proposition

DA outputs a stable matching.

Proof.

- Suppose $x - y$ is not in M.
- <u>Case 1:</u> y never proposed to x—then, y is matched to a woman he prefers to x.
- <u>Case 2:</u> y proposed to x—then, x rejected y and is matched to someone she prefers to y.
- In either case, $x - y$ is not a blocking pair.

$\square$

A Nobel Prize winning algorithm

Theorem

The DA algorithm terminates after $O(n^2)$ iterations and finds a stable matching for any problem instance.

History:

- Proposed by David Gale and Lloyd Shapley.
- Published in American Mathematical Monthly (1962).
- Nobel Prize in Economics (2012): Lloyd Shapley and Alvin Roth "for the theory of stable allocations and the practice of market design".

DA algorithm: practical applications

Education and School Choice

- University Admissions: Assigns students to universities based on mutual preferences, as in many centralized national admissions systems (e.g., UCAS in the UK).
- School Placement: Used by large school districts (New York, Boston) to allocate students to public schools, resolving complex preference lists and priorities.

Medical and Labor Markets

- Residency Matching: Forms the backbone of the National Resident Matching Program (NRMP) in the US, matching medical graduates to hospitals in a stable, incentive-compatible way.
- Other Labor Markets: Adapted for assigning new graduates to public service or academic positions.

DA algorithm: practical applications

Stable Marriage and Pairwise Matching

- Marriage Problem: Solves the classic stable marriage problem, forming stable matchings between two groups (e.g., mentors/mentees) when each has preferences over members of the other group.
- Adoption and Foster Care: Matches children to families considering preferences and priorities on both sides.

Resource and Capacity Allocation

- Hospital and School Capacity: Handles limited capacities—e.g., schools with quotas—so that each institution receives the number of agents it can serve while maintaining stability.
- Organ Allocation: Makes organ donor-recipient matches by incorporating preferences, priorities, and medical compatibility.
- Content delivery networks: Distributes much of world's content on the web: user preferences are based on latency and packet loss; web server preferences based on bandwidth and co-location.

Extension: many-to-one stable matching

Many-to-one matching:

- matching medical school students to hospitals: a hospital may be looking to hire several interns;
- matching children to schools: each school has several places.

Stability: there should be no pair $x - y$ where x is a school and y is a student, such that:

- y is not admitted to x;
- x prefers y to one of its current students;
- y prefers x to their current school.

The DA algorithm extends to this setting, producing a stable matching.

Extension: unacceptable pairs

- Some men/women (students/hospitals) may find some potential match unacceptable (e.g., a medical student may not want to move to Southampton).
- The numbers of men and women (students and slots) need not be equal.
- **Stability:** there should be no pair $x - y$ where x is a school and y is a student, such that:
  - y is not admitted to x;
  - y and x find each other acceptable;
  - x has a free slot or prefers y to one of its current students;
  - y is unmatched or prefers x to their current school.

The DA algorithm still works for this setting.

Extension: indifferences

Preferences may have *ties:*

- A school partitions students into 4 groups according to
  - whether they live in the catchment area (CA); and
  - whether they have a sibling in school (S).
- $\{CA, S\} \succ \{CA, \neg S\} \succ \{\neg CA, S\} \succ \{\neg CA, \neg S\}$
- Within each group, the school is indifferent.

(Weak) stability: there should be no pair $x - y$ where $x \in X$ and $y \in Y$, such that x and y strictly prefer each other to their current partners.

The DA algorithm still works for this setting.

THANK YOU!